

featuretree: Verified Recovery of Editable CAD Feature Trees from STEP, for Open CAD Systems

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<https://github.com/punkfab/featuretree>

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Abstract

Importing a STEP file into a parametric CAD system yields a single frozen solid: the design history is gone, and no pad depth or hole diameter can be edited. Recovering that history is a well-studied problem — the acceptance criterion we use was stated as a definition by Gupta and Nau [2] in 1995, and commercial implementations have shipped since roughly 1998. It is nonetheless absent from every open-source CAD system and from every callable library we could find: the capability exists only inside proprietary seats costing thousands of dollars per year, with no API. We describe **featuretree**, an MIT-licensed, training-free recogniser that recovers an editable feature tree from an exact B-rep, gates every recovery by re-executing the tree and comparing the result against the input, and emits into FreeCAD and Onshape — two systems with no import-time feature recognition at all. On the eleven-part NIST MBE PMI conformance corpus it returns VERIFIED for 3 parts, PARTIAL for 5, and explicitly refuses 3, producing *no silently incorrect trees*; all three VERIFIED recoveries exceed 99.68% IoU against the input. We also report a negative result about our own acceptance criterion: a scalar volume comparison lets over-cut and uncut material cancel, understating the true geometric discrepancy by $1.56\times$ on one part, which a two-sided IoU test eliminates. We claim no methodological novelty; we claim an available, honest implementation and a measured account of where it fails.

1 Introduction

A solid model in a neutral exchange format carries geometry but no design history. Opened in a parametric CAD tool it is one frozen body. Recovering an editable history from such a body is the classical problem of automated feature recognition (AFR), and it is genuinely old: Gupta and Nau [2] defined a recovered feature set as correct precisely when subtracting those features from stock reproduces the part, and Vandenbrande and Requicha [8] built a rule-based system that rejected invalid candidates and iterated to a complete decomposition. Both date from the mid-1990s.

The capability also ships. SolidWorks FeatureWorks has recognised features on imported bodies since approximately 1998, and PTC’s Creo Feature Recognition Tool, available since Pro/ENGINEER Wildfire 4.0, states in its documentation that it “validates the model to ensure that the new features do not modify its geometry” [5]. Geomagic Design X recovers a parametric tree and transfers it into five different CAD systems. The ISO 10303 standard went further still: Parts 55, 108, 111 and 112 standardise construction history, parameters and constraints, rolled into AP203 edition 2 and AP242 [4].

So why can you not do this? Every implementation above is a GUI feature inside a proprietary seat, priced from thousands to tens of thousands of dollars per year, and none exposes an API a program can call. The two CAD systems most accessible to an independent developer have nothing: Onshape’s documentation states that files imported from another system “do not have a feature tree in Onshape”, and FreeCAD has carried feature recognition as an open request since 2011, with maintainers reporting no capacity to implement it. The standardised route fared no better — Pratt and Kim [4] reported in 2006 that “there are at present no commercial STEP translators making use of” the construction-history capabilities, and we are not aware of published work since that revisits the question. Twenty years later the practical situation is unchanged.

This paper is a response to that gap, not to a gap in the methods literature.

Contributions.

1. **An open implementation (§3).** A training-free, MIT-licensed recogniser from exact B-rep to an editable feature tree, emitted natively into FreeCAD PartDesign and Onshape, and to a `build123d` solid, from one neutral intermediate representation. It is a library, not a GUI feature.
2. **An evaluation with an explicit refusal class (§4).** On the NIST MBE PMI conformance corpus we report VERIFIED/PARTIAL/REFUSED per part, with reconstruction IoU for every accepted recovery. We are not aware of another AFR evaluation that reports a refusal rate alongside a success rate.
3. **A negative result about the acceptance criterion (§5).** Over-cut and uncut material enter a volume difference with opposite signs. We measure the resulting cancellation and show why a two-sided test is required.

What this paper does not claim. It does not claim novelty for verification-gated recovery, for neutral feature representations, for multi-target emission, or for round-tripping GUI edits. Each is anticipated; §2 says by whom. The claim is availability, measurement, and honesty about failure.

2 Related work, and what it anticipates

We state the anticipating work plainly, because the contribution here does not depend on it being absent.

The acceptance criterion is not new. Gupta and Nau [2] require that subtracting the recovered features from stock reproduce the part exactly, and return a binary verdict — a part with no feasible plan “is considered unmachinable”. Ours is the same gate with a measured tolerance in place of set identity. In the learning literature, Willis et al. [9] accept a reconstruction when $\text{IoU} = 1$, noting explicitly that a recovered sequence need not match the ground-truth sequence: geometric equivalence rather than intent recovery, published in 2021.

Neutral feature representations are not new. The macro-parametric approach [3] defines tool-agnostic modelling commands replayed into a target system, and Proficiency’s commercial Universal Product Representation [6] held both parametric and B-rep versions of each feature and regenerated native trees in six CAD systems from around 2000.

Multi-target emission and GUI round-trip are not new. The BYU Neutral Parametric Canonical Form line [7] supports concurrent editing across NX, CATIA and Creo, capturing edits made in each GUI and propagating them through a neutral database. It keys on database GUIDs where we key on stable feature names — an implementation difference, not an inventive one.

Query-based references are not new. Cascaval et al. [1] formalise references as “queries that select elements from the constructed geometry”, exactly the alternative to stored kernel identifiers that we adopt, with correctness criteria we do not attempt.

What we did not find. We searched for, and did not find: any open-source B-rep-to-feature-tree recogniser; any AFR evaluation reporting a graded verdict with a measured geometric residual per part; and any published analysis of why the standardised AP242 construction-history route was not adopted. Absence in one search is weak evidence, and we report these as unfound rather than as nonexistent.

3 System

The intermediate representation is a list of named, ordered features — `sketch`, `pad`, `pocket`, `revolve`, `prism_cut`, `fillet` — as plain JSON. Geometry is selected by *query*, re-resolved against live geometry at each build rather than by persisted kernel identifiers, which is what lets one IR drive structurally different backends and lets edits be matched back by name. Authored IR uses human-meaningful names; recovered IR uses generated ones (`pocket0`, `cut2`), stable enough for the round trip but carrying no design intent.

Recovery proceeds cross-sectionally. Candidate axes are identified along which the solid is prismatic, or about which it is a body of revolution; the outline and through-holes are taken from cross-sections sampled at several heights; blind features are deferred rather than drilled speculatively, since over-cutting cannot be undone; and when the base extrude does not verify, the residual is carved feature by feature, each remaining lump itself recognised as a profile extruded along its own axis. Because several axes can appear prismatic, candidates are not scored heuristically — one is generated per axis and the verifier selects the winner.

The gate. Let S be the input solid, T a candidate tree and $E(T)$ its re-execution. We accept when the volume difference and a rotation-tolerant bounding-box comparison both fall within tolerance, and report PARTIAL with the residual otherwise. When no axis yields a recognisable extrude or body of revolution, the recogniser REFUSEDs outright rather than returning a best guess. Section 5 shows why this particular invariant is weaker than it looks.

4 Evaluation

Corpus. The NIST MBE PMI Validation and Conformance Testing artifacts, “AP203 geometry only” variants: eleven parts (CTC-01–05, FTC-06–11), pure B-rep with no PMI annotations. These are human-designed conformance artifacts intended to be difficult. NIST’s own documentation notes that they are not error-free reference files and that conformance checkers report syntax errors in some of them, so a refusal may reflect the input rather than the recogniser.

Table 1: Recovery over the eleven-part NIST corpus. IoU is computed after rotation-aware co-registration, because recovery works in the part’s own frame and the recovered solid is generally rotated relative to the input. Regenerate with `paper/figures/run_corpus.py` and `paper/figures/iou_corpus.py`.

Part	Verdict	$ \Delta V /V$ (%)	Features	Time (s)	IoU (%)
CTC-01	VERIFIED	0.10	51	15.7	99.841
CTC-02	REFUSED	—	—	66.4	—
CTC-03	PARTIAL	41.46	16	6.1	—
CTC-04	PARTIAL	2.43	270	37.2	—
CTC-05	PARTIAL	51.80	106	11.6	—
FTC-06	PARTIAL	31.45	41	8.7	—
FTC-07	REFUSED	—	—	8.4	—
FTC-08	PARTIAL	52.55	218	48.3	—
FTC-09	VERIFIED	0.19	80	5.1	99.684
FTC-10	REFUSED	—	—	8.5	—
FTC-11	VERIFIED	0.00	2	1.6	100.000

Results. Three parts VERIFIED (27%), five PARTIAL (45%), three REFUSED (27%). All three refusals were the same explicit rejection — no recognisable extrude or body of revolution — rather than a crash or a guess.

The result we consider most important is that **no incorrect tree was accepted**. Every VERIFIED recovery exceeds 99.68% IoU against the input. A 27% acceptance rate with an explicit refusal class is a different and more useful contract than a higher rate with no soundness signal, and it is the property that makes the output safe to edit.

Scale and comparability. This is eleven parts. It establishes that the verdict contract holds on non-trivial standard geometry; it does not establish a rate over a part distribution, and no general recovery rate should be read into it. FTC-11 in particular is a two-feature revolve recovered in 1.6s and should carry little weight. We note explicitly that these numbers are *not* comparable to the 99%-plus accuracies reported on MFCAD++ and similar benchmarks: those measure per-face classification against ground-truth labels on procedurally generated data, whereas this measures whether a recovered program rebuilds a human-designed conformance part.

5 A negative result: volume comparison admits cancellation

The scalar $|\Delta V|/V$ conceals its own structure. On CTC-01 the recovered solid is *smaller* than the original ($\Delta V = -14\,850\text{ mm}^3$), so the residual cannot be attributed to uncut chamfers, which would make it larger. Co-registering the solids and decomposing by inclusion–exclusion gives $19\,036\text{ mm}^3$ of *over-cut* against 4186 mm^3 of *uncut* material: opposite signs, partially cancelling, so the total geometric discrepancy (0.159 %) is 1.56 times the 0.101 % the verifier reports.

Nothing in a volume-plus-extent test bounds that cancellation factor. A VERIFIED verdict under it should be read as “reproduces the part’s volume and extent to within ϵ ”, not “is the part to within ϵ ”. A two-sided test — IoU, or a symmetric-difference bound — removes the failure mode entirely, because over-cut and uncut material both increase the symmetric difference. IoU is additionally the metric the CAD-program-inference literature already reports, so adopting it buys comparability at the same time. We report IoU in Table 1 for this reason, and recommend it as the acceptance criterion in place of the volume test the implementation currently ships.

A practical obstacle. The decomposition had to be derived from the union by inclusion–exclusion, because direct OpenCASCADE difference and intersection between these operands returned degenerate results — an empty intersection for solids whose bounding boxes coincide exactly. Any stronger verifier built on Booleans will meet the same problem.

6 Limitations

- **Eleven parts.** A demonstration of the contract, not a study of a rate. A corpus-scale evaluation remains the obvious next step.
- **partial failures are unquantified.** Our IoU harness pre-filters candidate orientations by bounding-box match, and a wrong recovery has a wrong bounding box, so the five PARTIAL parts have no IoU. We can say that they failed and by how much volume, but not how wrong they are geometrically.
- **Equivalence, not intent.** Verification certifies that $E(T)$ matches S , not that T is the designer’s history. Recovered feature names are generated and carry no design intent.
- **Edge blends are outside the vocabulary.** Fillets and chamfers are not swept profiles and are left as sub-tolerance residual, so a chamfered part can verify while the recovered tree has sharp edges.
- **Recovery is subtractive only.** Additive bosses cannot be recovered; such parts are reported PARTIAL, not approximated.
- **Freeform geometry is rejected, not approximated.** Splines, ellipses, lofts and sweeps admit no faithful sketch-and-pad tree.
- **Tolerances are uncalibrated.** ϵ was chosen to separate observed successes from observed failures on the parts tested. It is not derived from a manufacturing tolerance model.
- **Refusals are not attributed.** We did not separate refusals caused by geometry outside the vocabulary from refusals caused by defects in the input files, which NIST documents as present.

7 Conclusion

The methods here are not new; their availability is. A capability described in the literature in 1995, shipped commercially since 1998 and standardised in AP242 is still absent from open CAD, and that gap is a practical problem for anyone building on FreeCAD, Onshape or a code-CAD toolchain. We have shown that a training-free implementation reaches a useful contract on standard conformance geometry — accepting a quarter of the corpus with no incorrect acceptances, and refusing rather than guessing on the rest — and we have measured a real weakness in the acceptance criterion we began with, together with the two-sided replacement that removes it.

Availability

`featuretree` is MIT-licensed and has been publicly available since 2026-06-26 at <https://github.com/punkfab/featuretree>; the recognition component dates from 2026-07-10. Every number reported here is reproduced by the commands in `paper/figures/README.md`, verified against the repository state on 2026-09-19. The prior-art scan supporting §2 is in `paper/prior-art/FINDINGS.md`.

This paper is archived as a preprint at [10.5281/zenodo.22848793](https://zenodo.org/record/22848793). The software is archived separately: [10.5281/zenodo.22848722](https://zenodo.org/record/22848722) for version 0.1.0, or [10.5281/zenodo.22848721](https://zenodo.org/record/22848721) for the version-independent concept DOI, which always resolves to the latest release.

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